

Project Scenario

Business Problem:

A university currently registers students manually using paper forms and spreadsheets. The process is slow, errors are common, and students wait in long queues. The university wants an online Student Registration System.

Problem statement:

The university uses manual registration, causing long waiting times, duplicate records, and delayed course enrolment. The objective is to implement an online system that simplifies student registration processes and improves accuracy.

Business Objectives

- Reduce registration time
- Eliminate duplicate student records
- Allow students to register online
- Improves data accuracy
- Enable lectures to view class lists

Stakeholders and their responsibilities

Stakeholders	Roles
Students	Register for courses
System Admin	Approves registration
Lectures	View enrolled students
Finance Office	Confirms tuition payment
IT Department	Maintains the system

System Requirements

Functional Requirements

- ❖ Students can create an account
- ❖ Students can login
- ❖ Students can select modules
- ❖ Students can view their timetable
- ❖ Students can download proof of registration
- ❖ Registrar can approve registration
- ❖ Finance office can verify payment
- ❖ Lectures can view class lists

Non-Functional Requirements

- ❖ Login within 3 seconds
- ❖ Secure passwords
- ❖ Data backups everyday
- ❖ Mobile-friendly interface

System User Stories

- As a student I want to register for modules online so that I don't have to stand in long queues.
- As a student, I want to log in securely.
- As a registrar, I want to approve registrations.
- As a lecturer, I want to see my class list.
- As Finance office we want to verify payments.

As-Is Process (Current Process)

Student arrives on Campus



Collect paper form, from the admin office



Fill in the form manually



Administrator checks form



Finance verifies payment



Registration approved

To-Be Process (Future Process)

Student logs into the system



Completes registration online



Selects modules correctly



Payment verified automatically

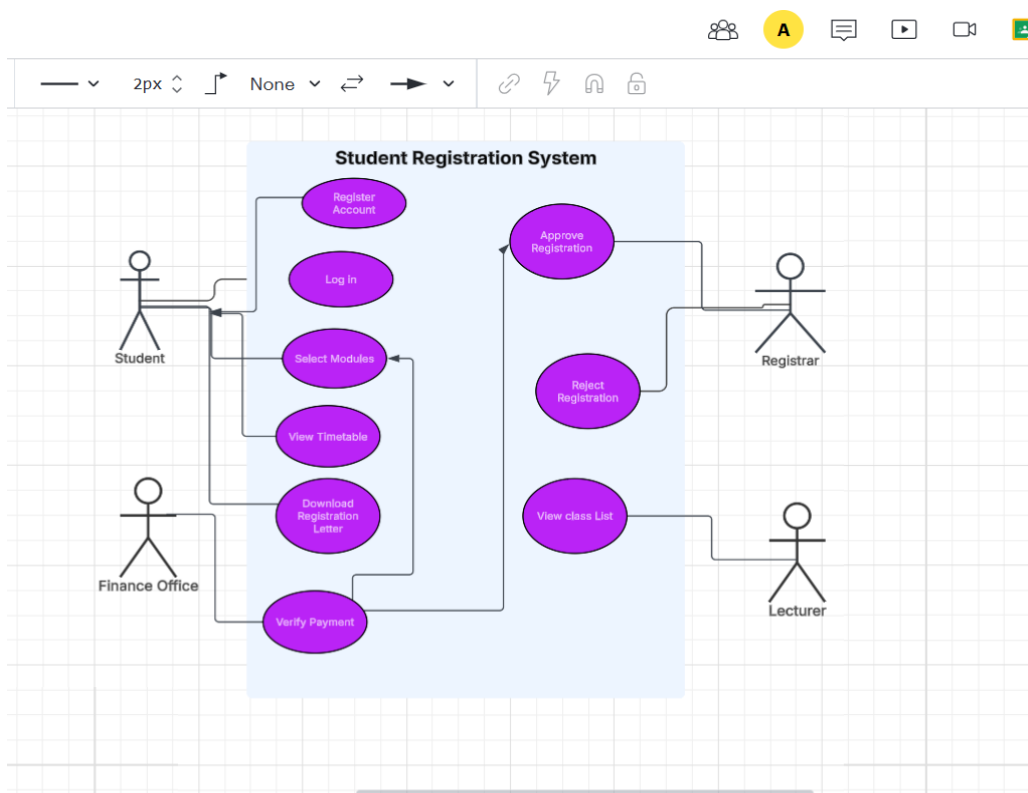


Registrar approves



Registration confirmation generated

Use Case Diagram



Acceptance Criteria

Register Account	• The student can enter all required personal information. • The system validates that required fields are completed. • The student receives a confirmation after successful registration. • The account is created successfully.
Log In	• The student enters a valid username and password. • The system grants access to the student dashboard. • Invalid credentials will display an error message.
Select Modules	• The student can view available modules. • The student can select one or more modules. • The system prevents duplicate module selection. • The selected modules are saved successfully.
Verify Payment	• Finance staff can verify the student's payment. • The system updates the payment status. • Registration cannot be approved if payment has not been verified.
Approve Registration	• The registrar can view pending registrations. • The registrar can approve or reject a registration. • The student is notified of the decision.
View Timetable	• The student can view their timetable after the registration is approved. • The timetable displays the correct modules and schedules.
Download Registration Letter	• The student can download a registration letter as a PDF. • The document includes the student's details and registered modules.
View Class Lists	• The lecturer can view the list of students enrolled in their modules. • The class list is up to date.

Potential Risks and Solutions

Risks	Solutions
➤ Students forget passwords	➤ Password reset feature
➤ System downtime	➤ Daily backups
➤ Incorrect payment information	➤ Automatic payment validation

Recommendations

1. Implement an online registration system to replace the manual paper-based process and reduce registration time.
2. Integrate the registration system with the finance system so that student payments can be verified automatically before registration is approved.
3. Send automated email or SMS notifications to students when their registration is approved or rejected.
4. Allow students to track their registration status (Pending, Approved, or Rejected) through their student portal.
5. Generate downloadable registration letters automatically after successful registration.
6. Implement role-based access control so that students, registrars, finance staff, and lecturers can only access features relevant to their roles.
7. Maintain regular data backups and security measures to protect student information and ensure system reliability.